

RESEARCH ARTICLE

A Thermomechanical Model of Coarse-Grained Soils With Non-Orthogonal Flow Rule

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ABSTRACT

In geo-energy engineering projects, temperature is an essential environmental variable, and accurately predicting its effect on the thermomechanical properties of geomaterials remains a challenge. Similar to fine-grained soils, temperature variation is also a crucial factor that affects the stress-strain response and critical state behavior of coarse-grained soils. In this study, a thermomechanical model is established for coarse-grained soils using theories from the critical state and fractional plasticity. The evolution of the critical state line with increasing temperature can be well characterized by introducing a thermal-dependent parameter, then the state void-ratio-pressure parameter that incorporates the effect of temperature can be derived according to the temperature-dependent critical state. The plastic flow direction and dilatancy function are obtained directly from the fractional derivation of the modified elliptical yield function to describe the nonassociated flow characteristics. Furthermore, the thermal state parameter is introduced into the non-orthogonal dilatancy function and hardening modulus to reflect the state- and temperature-dependent behaviors. Comparative analysis of experimental data and predictions indicates that the established thermomechanical model can reasonably predict the drained shear characteristics of coarse-grained soils under different temperatures, including strain-hardening, strain-softening, dilatancy, contraction, and thermal softening.

1 | Introduction

In geo-energy engineering projects such as radioactive waste disposal [1, 2], energy piles [3, 4], geothermal pavement [5, 6], heated pipelines [7, 8], and landfill treatment [9], geomaterials are constantly subjected to nonisothermal conditions. In recent years, many scholars have conducted in-depth research on different types of soils through experimental methods, including clay, silt, sand, and so forth. Investigations on the thermomechanical

behaviors of coarse-grained soils have been scarcer than those on fine-grained soils [10–14], but recent experimental results have confirmed that temperature fluctuations can cause permanent changes in the microporous structures and mechanical properties of coarse-grained soils [15–18]. Therefore, to comprehensively explain and predict the thermomechanical behavior of coarse-grained soils, the establishment of an applicable and reliable constitutive model has important scientific and engineering value.

In recent decades, various constitutive models have been constructed to reproduce the complex stress-strain behaviors of coarse-grained soils under different theoretical frameworks, such as bounding surface theory [19–21], the subloading surface concept [22], the unified hardening concept [23, 24], generalized plasticity theory [25, 26], and hypoplastic theory [27–30]. Nevertheless, the aforementioned models are incapable of predicting the thermoplastic behavior of coarse-grained soils. For fine-grained soils, temperature-dependent yield surfaces are typically used to describe thermomechanical behaviors [31–37]. The non-isothermal plasticity model proposed by Laloui and Cekerevac [38] accounts for the influence of temperature on the yield stress and friction angle. Yao and Zhou [39] noted that as the temperature increased, the yield stress of soils decreased and the yield surface decreased. Abuel-Naga et al. [40] reported that temperature effects induced anisotropy in soft Bangkok clay, and introduced an additional parameter to modify the yield surface under different temperature conditions. Two yield surfaces and loading yield limits were used by Cheng et al. [41] to construct a thermomechanical model considering the thermal cyclic behavior of fine-grained soils. Zhu et al. [42] adopted a temperature-dependent state parameter and introduced two coupled yield mechanisms to present a thermomechanical constitutive model. Based on thermodynamic principles, a novel temperature-dependent dissipation potential was derived by Golchin et al. [43]. Then a thermomechanical model of fine-grained soils was established by considering the implications of thermal volume and shear behavior.

The deformation of coarse-grained soil has obvious nonassociated flow characteristics [44–47]. The associated and nonassociated flow rules depend on whether the plastic potential function (PPF) is consistent with the yield function (YF). To obtain the purpose of nonassociated flow, a new PPF needs to be constructed [48–50]. However, it is more difficult to obtain the required PPF for geomaterials because of dilatancy. Since fractional derivatives exhibit the capability to control the non-orthogonal gradient direction of differentiable functions, they are considered an important approach for obtaining the plastic flow laws that uniformly describe the associated and nonassociated of geomaterials. By utilizing the non-orthogonal gradient of the YF, the direction of plastic flow was determined by Lu et al. [51], and the tensor expression of the plastic strain increment was further deduced. The non-orthogonal plastic flow rule provides a new concept and approach for describing the non-orthogonality between the incremental direction of plastic strain and the YF in the stress space, avoiding the difficult problem of constructing the PPF. The advantages of this solution in describing the plastic deformation of materials are widely recognized and have been applied in the constitutive modeling of different materials, including granular soil [52–54], clay [50, 55–57], rock [58, 59], and concrete [60–62].

In this context, this research intends to develop a thermomechanical model of coarse-grained soils using theories from the critical state and fractional plasticity. The direction of the non-orthogonal plastic flow is directly obtained by the fractional derivative of the modified elliptical YF. Furthermore, considering the key feature of the thermally induced shear behavior, that is, the critical state line (CSL) moves with the variation in temperature, a state variable that can simultaneously reflect the effects of temperature, pressure, and density is incorporated. The

thermal state parameter is introduced into the non-orthogonal dilatancy function and hardening modulus to reflect the state- and temperature-dependent behaviors. Finally, the rationality and accuracy of the state dependent non-orthogonal thermo-mechanical (SDNTM) model are verified against a series of temperature-controlled triaxial drained shear (TCTDS) test data collected from different studies.

2 | Model Establishment

2.1 | Temperature-Dependent Critical State

Experimental evidence reported in the literature has demonstrated that the location of the CSL in the e - p (i.e., void ratio—mean effective stress) plane is dramatically affected by temperature fluctuations [12, 16, 63, 64]. Figure 1 shows the CS points of Fujian sand and silty phyllite in the e - $(p/p_a)^\xi$ plane at different temperatures. The positions of the CSP in different stress states are approximately parallel downward with increasing temperature T . That is, the CSL has a significant temperature dependence. For granular soils, the expression of the CSL in the compression plane can be written as:

$$e_{cs} = \Gamma(T) - \lambda(p/p_a)^\xi \quad (1)$$

where $\Gamma(T)$ and λ are the intercept and slope of the CSL, respectively; and ξ is a material parameter, that varies between 0.6 and 0.8.

As depicted in Figure 1, the correlation between the intercepts of the CSLs and temperature needs further consideration to capture the thermal effects on the shear behavior of coarse-grained soils in a more rational way. Thus, the relationship between parameters $\Gamma(T)$ and T can be quantified by a logarithmic function [65]:

$$\Gamma(T) = \Gamma_0 - \vartheta_T \ln(T/T_0) \quad (2)$$

where Γ_0 denotes the intercept at T_0 and ϑ_T is a temperature-dependent parameter.

For coarse-grained soils, the pressure and density of the specimen often vary within a wide range, which has distinct state dependent characteristics. The state parameter ψ ($= e - e_{cs}$), as the most commonly used variable, only accounts for the variation in the void ratio, which has some limitations in practical applications. In this study, the state void-ratio-pressure parameter I_{ep} proposed by Xiao and Liu [66] is applied (as shown in Figure 2), as follows:

$$I_{ep} = \frac{e}{e_{cs}} \frac{p}{p_{cs}} \quad (3)$$

By substituting Equations (1) and (2) into Equation (3), the thermal state void-ratio-pressure parameter can be rewritten as:

$$I_{ep} = \frac{ep / (\Gamma_0 - \vartheta_T \ln(T/T_0) - \lambda(p/p_a)^\xi)}{p_a((\Gamma_0 - e - \vartheta_T \ln(T/T_0)) / \lambda)^{1/\xi}} \quad (4)$$

Based on experimental observations, temperature variations also affect the CSL in the q - p plane (q denotes the deviatoric stress)

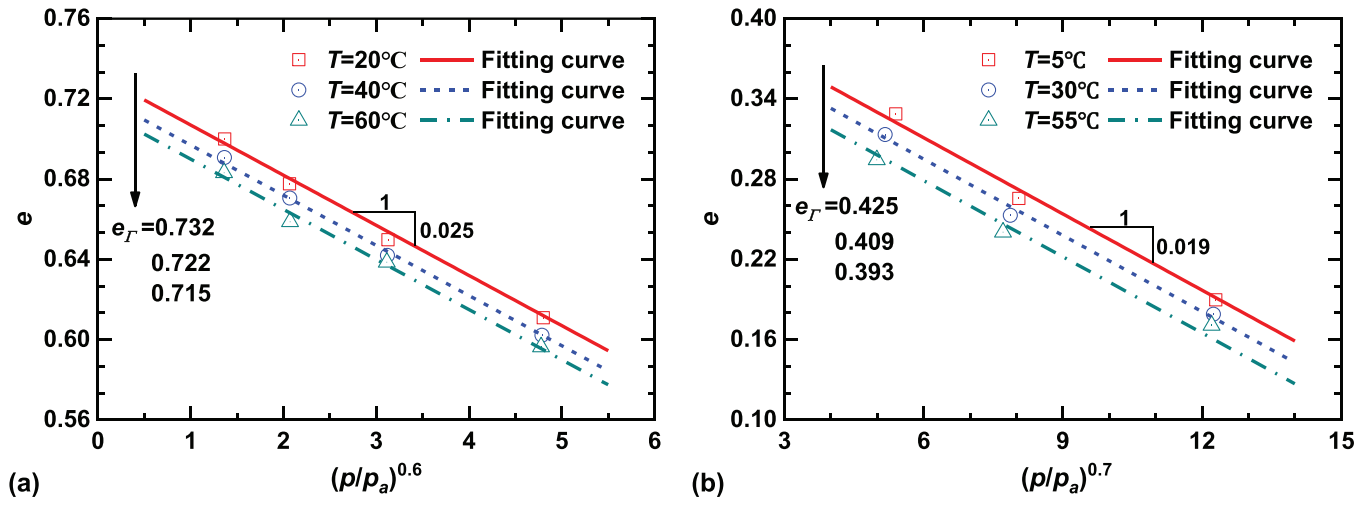


FIGURE 1 | CSLs in the $e - (p/p_a)^\xi$ at different temperatures: (a) Fujian sand [63]; (b) silty phyllite [64].

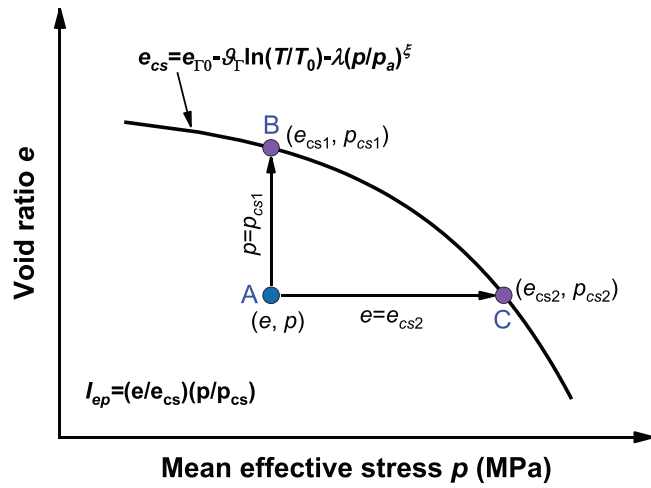


FIGURE 2 | Definition of the state void-ratio-pressure parameter.

[13, 16]. At high temperatures, the critical state stress ratio M may decrease, increase or remain unchanged [67]. Hence, the linear function with temperature can be expressed as:

$$M_T = M - \vartheta_T (T - T_0) \quad (5)$$

where ϑ_T is a temperature dependent material parameter.

2.2 | Yield Surface

The shape of the yield surface (YS) adopts the modified elliptical surface proposed by Bardet [68], as shown in Figure 3, which can be defined as follows:

$$f_T = \frac{(p - \chi p_{cT})^2}{(1 - \chi)^2 p_{cT}^2} + \frac{q^2}{(M_T \chi p_{cT})^2} - 1 = 0 \quad (6)$$

where χ is the elliptical aspect ratio, which controls the shape of elliptical; p_{cT} is the intersection point of the YS with the p -axis at temperature T . A thermal yielding curve was introduced by Cui et al. [32] with an exponential expression as follows, $p_{cT} =$

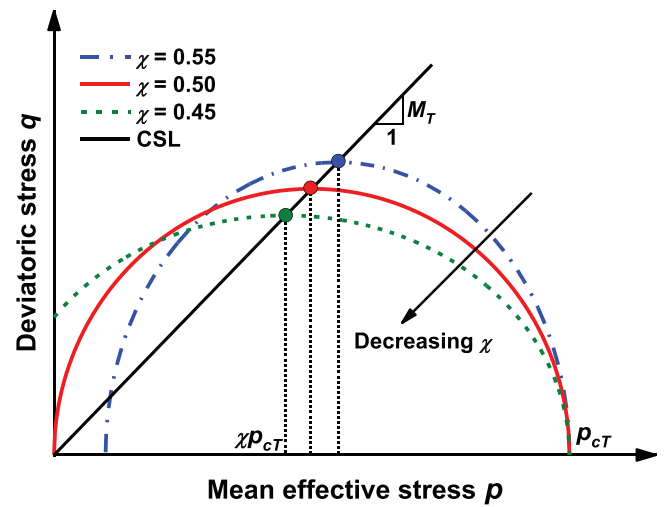


FIGURE 3 | Schematic diagram of yield surfaces with different values of χ .

$p_{cT0} \exp[-\beta(T - T_0)]$, where p_{cT0} is the yield stress at T_0 ; and β is a material parameter. Through M_T and β , the YS (i.e., size and shape) gradually changes at elevated temperatures.

Differential Equation (6) can be obtained as follows:

$$\frac{\partial f_T}{\partial p} = 2(\rho - \chi) \chi^2 M_T^2 p_{cT} \quad (7)$$

$$\frac{\partial f_T}{\partial q} = 2(1 - \chi)^2 \rho \eta p_{cT} \quad (8)$$

$$\begin{aligned} \frac{\partial f_T}{\partial T} &= \frac{\partial f_T}{\partial p_{cT}} \frac{\partial p_{cT}}{\partial T} + \frac{\partial f_T}{\partial M_T} \frac{\partial M_T}{\partial T} = 2\beta[\chi(\rho - \chi) \\ &+ (1 - \chi)^2 M_T^2 \chi^2 p_{cT}^2 - 2\vartheta_T[(\rho - \chi)^2 - (1 - \beta)^2] M_T p_{cT}^2 \chi^2] \end{aligned} \quad (9)$$

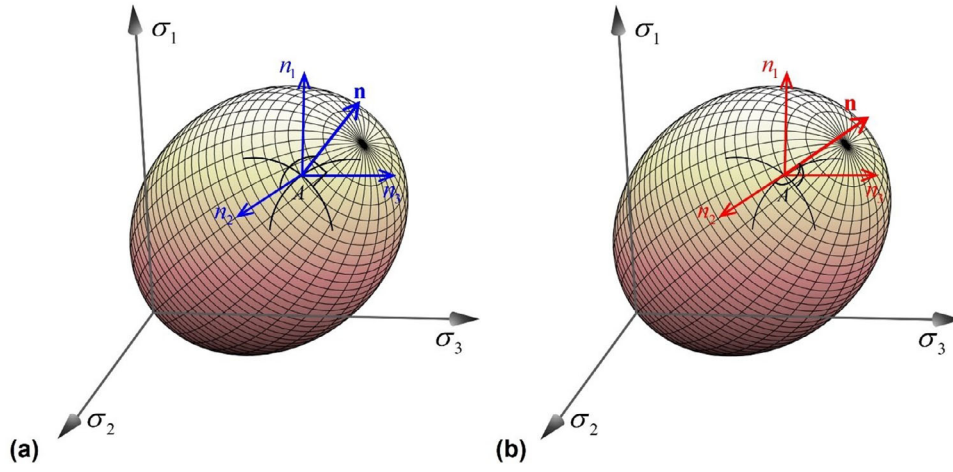


FIGURE 4 | Yield surface gradient vectors under different fractional orders [60]: (a) $\alpha = 1$; (b) $\alpha \neq 1$.

where $\eta (= q/p)$ is the stress ratio. The scalar ρ is obtained as follows:

$$\rho = \frac{M_T^2 \chi^3 + \sqrt{M_T^2 (\chi - 1)^2 \chi^2 ((1 - 2\chi)\eta^2 + M_T^2 \chi^2)}}{M_T^2 \chi^2 + (\chi - 1)^2 \eta^2} \quad (10)$$

2.3 | Non-Orthogonal Flow Rule

Fractional calculus is widely used in many fields, and various definitions have been developed, such as the Caputo, Riemann-Liouville, and Grünwald-Letnikov derivatives [55, 69, 70]. The main difference between the Caputo and Riemann-Liouville definitions is the order of derivation and integration. Although the definition of Caputo derivatives is simple and widely used, it is unstable for elliptical YF adopted in this work when the fractional order is greater than 1 because the differential of constant is 0. Therefore, the Riemann-Liouville fractional derivative is adopted in reference to the work of Lu et al. [60], as follows:

$${}^{RL}_a D_t^\alpha f(t) = \frac{1}{\Gamma(n - \alpha)} \frac{d^n}{dt^n} \left(\int_a^t (t - \tau)^{n-\alpha-1} f(\tau) d\tau \right) \quad (11)$$

where ${}^{RL}_a D^\alpha$ denotes the Riemann-Liouville fractional derivative of order α ; a is the starting point of the independent variable t ; n is the minimal positive integer, $\alpha \in (n - 1, n]$; $\Gamma(\cdot)$ denotes the gamma function ($\Gamma(z) = \int_0^\infty e^{-\tau} \tau^{z-1} d\tau$, $\text{Re}(z) > 0$); and $f^{(n)}(\tau)$ is the n -order derivative of $f(\tau)$. Figure 4 schematically illustrates the yield surface and gradient vectors when the fractional order $\alpha = 1$ and $\alpha \neq 1$.

Substituting Equation (6) into (10), the fractional derivatives of the YS can be written as:

$$\begin{aligned} {}^{RL}_0 D_p^\alpha f_T(p) &= \frac{\partial^\alpha f_T}{\partial p^\alpha} = \frac{[2(1 - \alpha)(2 - \alpha)(1 - \chi)] M_T^2 \chi^3}{\rho^\alpha p_{cT}^{\alpha-2} \Gamma(3 - \alpha)} \\ &+ \frac{2M_T^2 \chi^2 [\rho - (2 - \alpha)\chi]}{\rho^{\alpha-1} p_{cT}^{\alpha-2} \Gamma(3 - \alpha)} + \frac{(1 - \alpha)(2 - \alpha)(1 - \chi)^2}{\eta^2 \rho^{\alpha-2} p_{cT}^{\alpha-2} \Gamma(3 - \alpha)} \end{aligned} \quad (12)$$

$$\begin{aligned} {}^{RL}_0 D_q^\alpha f_T(q) &= \frac{\partial^\alpha f_T}{\partial q^\alpha} = \frac{2(1 - \chi)^2 \eta^{2-\alpha}}{\rho^{\alpha-2} p_{cT}^{\alpha-2} \Gamma(3 - \alpha)} \\ &+ \frac{M_T^2 \chi^2 (\rho^2 - 2\chi\rho + 2\chi - 1)(1 - \alpha)(2 - \alpha)}{\eta^\alpha \rho^\alpha p_{cT}^{\alpha-2} \Gamma(3 - \alpha)} \end{aligned} \quad (13)$$

Then, the fractional order dilatancy function D is derived as:

$$D = -\frac{d\epsilon_v^p}{d\epsilon_s^p} = \frac{\partial^\alpha f_T / \partial p^\alpha}{\partial^\alpha f_T / \partial q^\alpha} \quad (14)$$

Combining Equations (12), (13), and (14) yields:

$$\begin{aligned} D &= \left\{ 2M_T^2 \chi^2 \rho [\rho - (2 - \alpha)\chi] + \right. \\ &(2 - 3\alpha + \alpha^2) \left[(2 - \chi) M_T^2 \chi^3 + (1 - \chi)^2 \eta^2 \chi^2 \right] \Big\} / \\ &\left\{ M_T^2 \chi^2 (\rho^2 - 2\chi\rho + 2\chi - 1)(2 - 3\alpha + \alpha^2) \eta^{-\alpha} \right. \\ &\left. + 2(1 - \chi)^2 \eta^{2-\alpha} \rho^2 \right\} \end{aligned} \quad (15)$$

When the fractional order $\alpha=1.0$, the dilatancy equation of Equation (15) can be simplified as:

$$D_{\alpha=1} = \frac{M_T^2 \chi^2 (\rho - \chi)}{(1 - \chi)^2 \rho \eta} = \frac{\partial f_T / \partial p}{\partial f_T / \partial q} \quad (16)$$

The fractional order α is associated with the state variable, which varies with density and stress state of coarse-grained soils. Thus, the thermal state void-ratio-pressure parameter I_{ep} is introduced into the fractional order in this study [53, 71, 72]:

$$\alpha = 1 + m \ln I_{ep} \quad (17)$$

where m is a parameter. When the stress reaches the CS, rearranging Equation (15) shows that the dilatancy D satisfies the condition $D|_{\eta=M_T, \alpha=1} = 0$.

2.4 | Hardening Modulus

There are two approaches to determine the hardening modulus (H_p): one is strictly adhere to the consistency condition of the

YF; the second is to construct a mathematical expression that satisfies the following conditions: (i) initial state $\eta=0, H_p = +\infty$; (ii) critical state $\eta = M_T$ and $e = e_{cs}, H_p = 0$; and (iii) peak state $\eta = M_p, H_p = 0$. Referring to previous works [73–75], a state- and temperature-dependent H_p is defined as:

$$H_p = h_0 G p \left(1 - \frac{I_{ep}}{2} \right) \frac{M_p - \eta}{\eta(1 + \eta)^2} \quad (18)$$

where G is the elastic shear-modulus. M_p denotes the peak-failure stress ratio, and an expression with respect to the thermal state parameters can be defined as follows:

$$M_p = M_T (1 - k_p \ln I_{ep}) \quad (19)$$

where k_p is a parameter pertaining to the peak-failure strength. For the loose state, the current stress point is above the CSL, $I_{ep} > 1$, $\alpha > 1$, and $M_p < M_T$; for the dense state, the current stress point is below the CSL, $I_{ep} < 1$, $\alpha < 1$, and $M_p > M_T$; and when the current stress point is on the CSL, $I_{ep} = 1$, $\alpha = 1$, and $M_p = M_T$. Thus, I_{ep} further changes the stress-strain response of soils by adjusting the dilatancy and peak-failure strength.

2.5 | Constitutive Equations

It is usually assumed that the temperature variation only causes elastic volumetric strain, and the elastic deviatoric strain is independent of temperature [31, 34, 39]. The total strain increment can be divided into three parts: elastic strain increment ($d\epsilon_v^e$, $d\epsilon_s^e$), thermoelastic volumetric strain increment ($d\epsilon_v^T$), and plastic strain increment ($d\epsilon_v^p$, $d\epsilon_s^p$), in which $d\epsilon_v^e$, $d\epsilon_s^e$, and $d\epsilon_v^T$ can be expressed as:

$$d\epsilon_v^e = \frac{1}{K} dp \quad (20)$$

$$d\epsilon_s^e = \frac{1}{3G} dq \quad (21)$$

$$d\epsilon_v^T = -\alpha_T dT \quad (22)$$

where α_T is the thermal expansion coefficient of the solid skeleton. The elastic shear-modulus G and bulk-modulus K are obtained as follows:

$$G = G_0 p_a \frac{(2.97 - e)^2}{1 + e} \sqrt{\frac{p}{p_a}} \quad (23)$$

$$K = \frac{2(1 + \nu)}{3(1 - 2\nu)} G \quad (24)$$

where G_0 is a material constant; and ν is Poisson's ratio.

Based on the non-orthogonal flow rule, the plastic strains $d\epsilon_v^p$ and $d\epsilon_s^p$ are as follows:

$$d\epsilon_v^p = \frac{1}{H_p} \left(\frac{\partial f_T}{\partial p} dp + \frac{\partial f_T}{\partial q} dq + \frac{\partial f_T}{\partial T} dT \right) D \quad (25)$$

$$d\epsilon_s^p = \frac{1}{H_p} \left(\frac{\partial f_T}{\partial p} dp + \frac{\partial f_T}{\partial q} dq + \frac{\partial f_T}{\partial T} dT \right) \quad (26)$$

TABLE 1 | Model parameters.

Symbols	Fujian sand	Silty phyllite rockfill	Crushed tuff
G_0	128.0	120.0	96.0
ν	0.25	0.25	0.25
α_T	0.00003	0.00003	0.00003
M	1.252	1.775	1.905
ϑ_T	0.0003	0.0012	0.0
Γ_0	0.732	0.426	0.415
ϑ_Γ	0.015	0.013	0.008
λ	0.025	0.019	0.031
ξ	0.6	0.7	0.6
χ	0.55	0.52	0.53
m	0.03	0.28	0.05
ζ_p	0.13	0.32	0.23
β	0.002	0.0015	0.0012
h_0	0.9	1.4	2.8

The thermomechanical model in this study considers the influence of temperature on the CSL and state-dependent behavior of coarse-grained soils. Notably, the proposed model also has some limitations that need to be improved upon in future research, such as ignoring the impact of particle breakage and energy non-conservation in the elastic region [76, 77].

3 | Determination of the Model Parameters

The thermomechanical model for coarse-grained soils with non-orthogonal flow rule requires 14 parameters: (i) three elastic parameters G_0 , ν , and α_T ; (ii) six critical state parameters M , ϑ_T , Γ_0 , ϑ_Γ , λ , and ξ ; (iii) two dilatancy parameters χ and m ; (iv) two strength parameters ζ_p and β ; and (v) one plastic modulus parameter h_0 . It is generally believed that the soil deformation is still in the elastic deformation stage within the range of $\epsilon_a \leq 0.2\%$, and the initial shear-modulus G_0 can be calculated from the $q - \epsilon_s$ curves by the following equation [66]:

$$G_0 = \frac{dq}{3d\epsilon_s} \frac{(1 + e)}{(2.97 - e)^2 \sqrt{p p_a}} \quad (27)$$

The determination procedure of parameter ν can be found in previous studies, and a constant value of 0.25 is used for coarse-grained soils. The parameter α_T can be obtained from the heating-cooling test [39], and owing to the lack of test data, a value of 0.00003 is uniformly taken. Parameters M and ϑ_T can be determined from the CS data at different temperatures in the $q - p$ plane with Equation (5). In the $e - p$ plane, Γ_0 , ϑ_Γ , λ , and ξ can be calculated from the CS data at different temperatures with Equations (1) and (2), as shown in Figure 1. The shape parameter χ controls the dilatancy of soils, and the value of χ can be determined by adjusting the size of χ to obtain the best fitting result of the test data in the $D - \eta$ plane [19, 47]. The fractional order α reflects the degree of nonassociated flow of geomaterials.

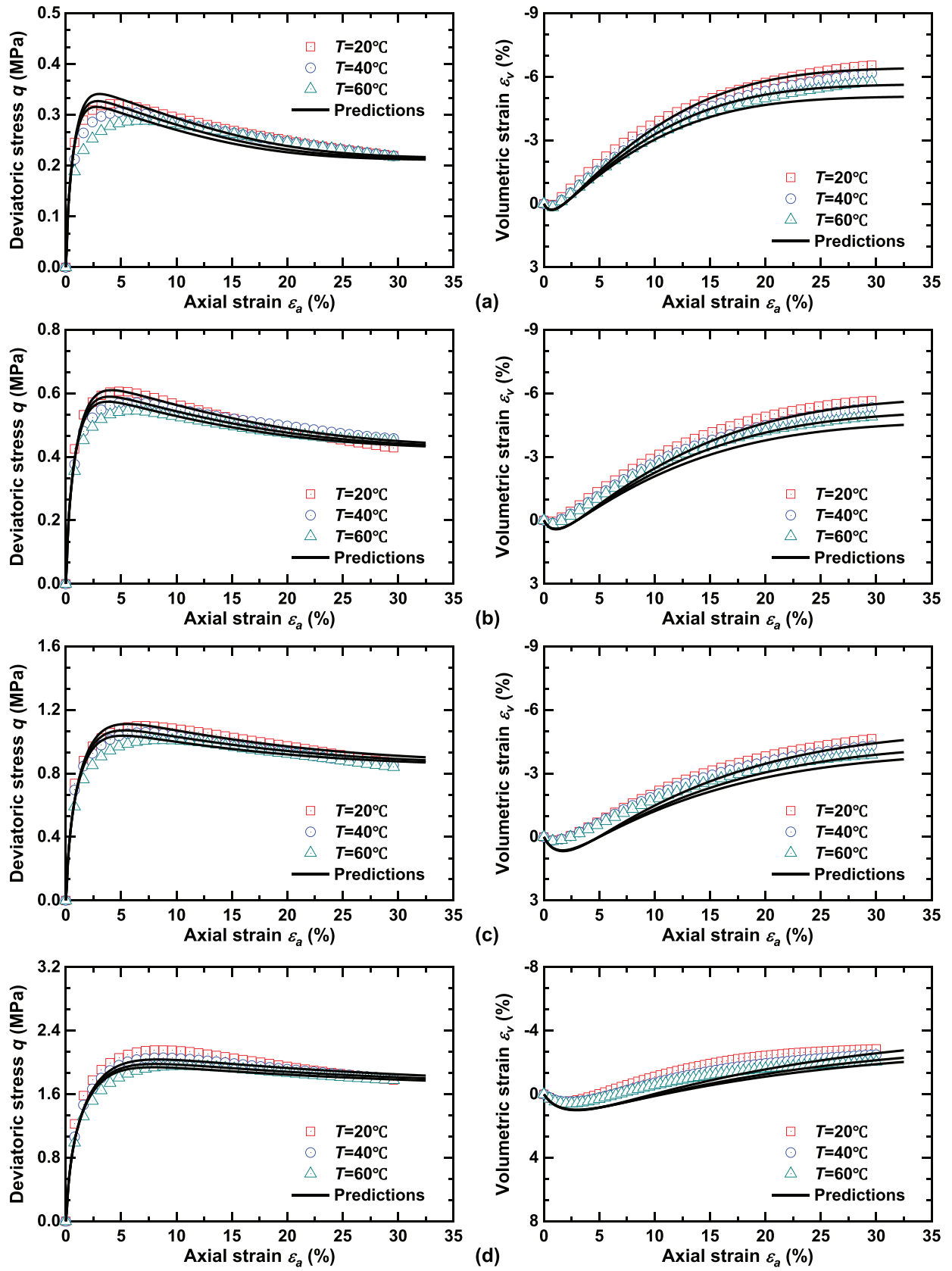


FIGURE 5 | Model predictions versus test data of TCTDC tests on Fujian sand: (a) $p_c = 0.1$ MPa; (b) $p_c = 0.2$ MPa; (c) $p_c = 0.4$ MPa; (d) $p_c = 0.8$ MPa.

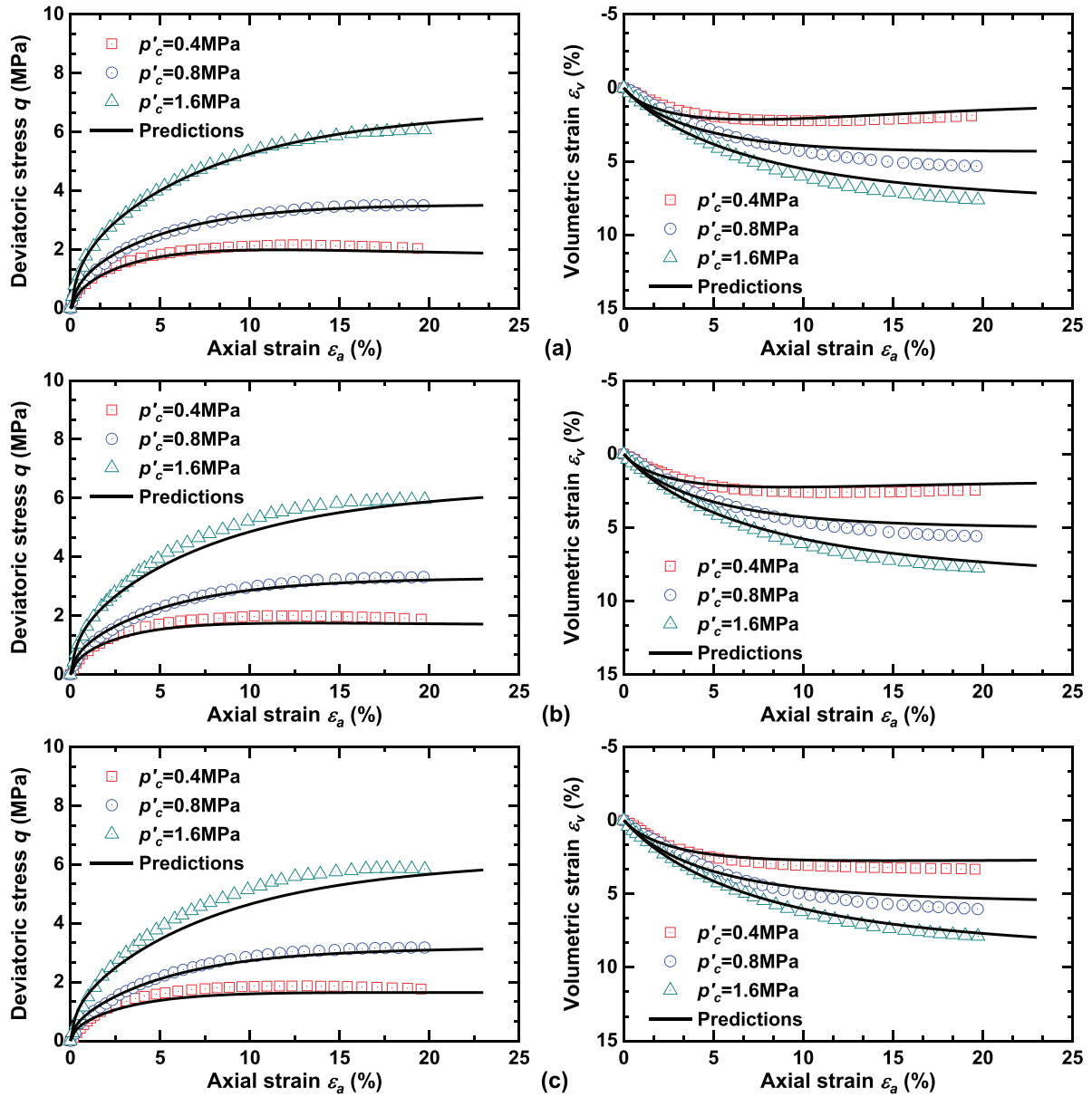


FIGURE 6 | Model predictions versus test data of TCTDC tests on silty phyllite rockfill: (a) $T = 5^\circ\text{C}$; (b) $T = 30^\circ\text{C}$; (c) $T = 55^\circ\text{C}$.

At a phase transformation state $D = 0$, the dilatancy parameter m can be estimated by combining Equations (16) and (18) [54, 72]. At a peak-failure state ($\eta = M_p$, $H_p = 0$), thus parameter k_p can be obtained as follows:

$$k_p = \frac{M_T - (M_p)_p^{ob}}{M_T \ln(I_{ep})_p^{ob}} \quad (28)$$

where $(I_{ep})_p^{ob}$ and $(M_p)_p^{ob}$ are the observed values of I_{ep} and M_p at the peak-failure state, respectively. Then, Equation (8) can be used to calculate parameter β when the yield stress p_{cT} is obtained from the temperature-controlled isotropic consolidation test [78]. For the drained triaxial test ($dp' = dq/3$), h_0 can be obtained from the $q - \varepsilon_s$ curve, as follows:

$$h_0 = \frac{\eta(1+\eta)^2 \left(\frac{1}{3} \frac{\partial f_T}{\partial p} + \frac{\partial f_T}{\partial q} \right) \frac{\partial^\alpha f_T}{\partial q^\alpha}}{Gp(M_p - \eta) \left(\frac{d\varepsilon_s}{dq} - \frac{1}{3G_e} \right)} \quad (29)$$

4 | Model Validation

To evaluate the validity and capability of the proposed SDNTM model, a series of TCTDC test data for Fujian sand, silty phyllite rockfill (SPR), and crushed tuff are selected from the literature. The parameters used for model validation are summarized in Table 1.

4.1 | Fujian Sand

Xiao et al. [79] conducted plenty TCTDC tests to explore the thermal effects on the shear response of Fujian sand. The particle sizes of Fujian sand range from 0.15 to 1.2 mm. The specimens were first consolidated at pressures of 0.1, 0.2, 0.4, and 0.8 MPa, and then gradually heated from 20°C to the target temperature (40°C and 60°C) under constant pressure. The initial void ratios under different confining pressures given above are 0.595, 0.585,

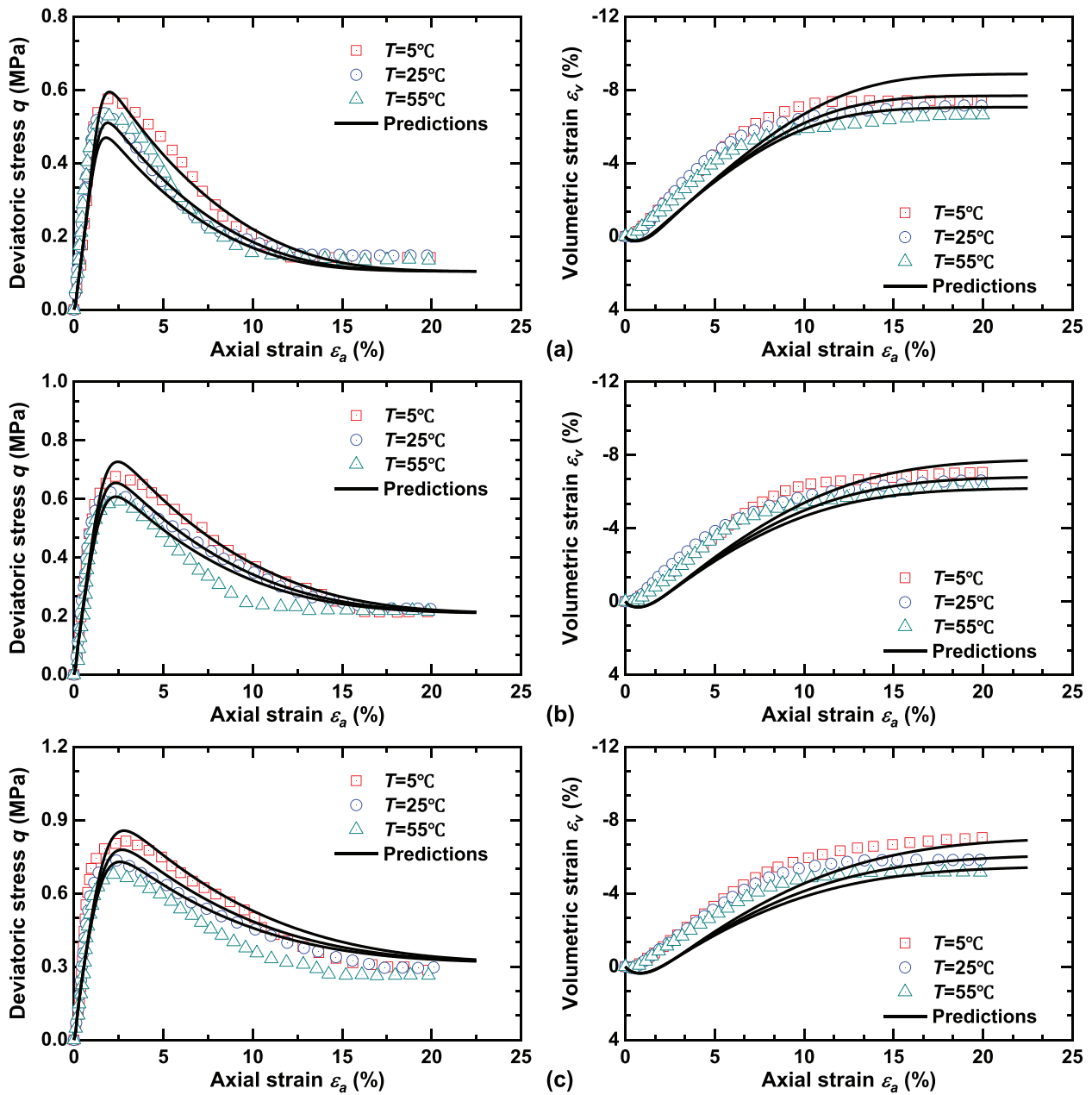


FIGURE 7 | Model predictions versus test data of TCTDC tests on crushed tuff: (a) $p_c = 0.02$ MPa; (b) $p_c = 0.04$ MPa; (c) $p_c = 0.06$ MPa.

0.577, and 0.560, respectively. The TCTDC tests were performed at $T = 20^\circ\text{C}$, 40°C , and 60°C when there were no obvious fluctuations in temperature. The stress-strain response of Fujian sand is significantly affected by temperature variations, as shown in Figure 5. As the temperature increases, the peak-state shear strength of the specimen decreases, exhibiting the phenomenon of thermal softening. In addition, it is observed that the specimens at high temperatures exhibit more volumetric contraction than those at low temperatures. Figure 5 presents comparisons between the model predictions and TCTDC test results for Fujian sand. The model can provide accurate predictions for deviatoric stress and volumetric strain at various p_c and T values, and the post-peak strain softening behavior for low p_c specimens can also be reasonably predicted. That is, the proposed thermomechanical model can adequately capture the strain-softening, dilatancy, and thermal softening behaviors of Fujian sand.

4.2 | Silty Phyllite Rockfill

SPR is a kind of coarse-grained material commonly used in engineering, and has the characteristics of a high degree of weathering, low compressive strength and high sensitive to temperature. To investigate the temperature sensitivity of shear strength, shear deformation, and critical state behavior, large-scale TCTDC tests of SPR were carried out by Mao et al. [64]. The specific gravity of the SPR is 2.64, and the dry water absorption of the saturated surface is 1.78%. The size of the specimen was $200\text{ mm} \times 400\text{ mm}$, and the maximum particle size was $d_{\max} = 40\text{ mm}$. Three different temperatures were set at 5°C , 30°C , and 55°C , and the specimens under different T conditions were subjected to $p_c = 0.4$, 0.8 , and 1.6 MPa for the large-scale TCTDC tests. The initial void ratios under these different confining pressures are 0.350, 0.335, and 0.288, respectively.

The mechanical properties of the SPR have obvious thermal effects, with increasing T , the strength decreases and the shear deformation enhances. The influence of temperature on the stress-strain curves of the SPR is more obvious at low p_c , and the difference caused by T gradually decreases with increasing p_c , as shown in Figure 6. It can be found that an increase in p_c , the transition of the volumetric strain from contraction to expansion for three temperatures can be effectively captured. The predicted $q - \varepsilon_a$ and $\varepsilon_v - \varepsilon_a$ curves of the thermomechanical model agree well with the large-scale TCTDC test results. The comparison shows that the proposed SDNTM model can reasonably simulate the thermomechanical behavior of the SPR under triaxial drained conditions.

4.3 | Crushed Tuff

Cao et al. [80] conducted large-scale TCTDC tests to explore the thermally-induced shear responses of crushed tuff at various confining pressures and temperatures. The crushed tuff was classified as well-graded gravel, with an average particle size of 5.76 mm, a coefficient of uniformity of 22.7, and coefficient of curvature of 2.88. The compaction value of the specimen was 95%, and the initial void ratio was $e_0 = 0.283$. To reflect the thermally-induced shear effects, the differences in the static characteristics of coarse-grained soil at high, room, and low temperatures (i.e., 55°C, 25°C, and 5°C) were compared. Meanwhile, three conditions of low confining pressure (0.02, 0.04, and 0.06 MPa) consider the low confining stress state of the subgrade fill. Figure 7 displays the stress-strain relationships for crushed tuff at different temperatures. The simulation results can reasonably reflect the experimental phenomenon, that is, the peak strength and dilatancy decrease with increasing T , resulting in thermal softening. In addition, the higher p_c is, the more pronounced the peak strength attenuation, but the temperature variation has no effect on the residual strength of the crushed tuff. Overall, the proposed thermomechanical model can accurately reflect the shear characteristics at different temperatures.

5 | Conclusions

To depict the thermomechanical behavior of coarse-grained soils more comprehensively, a thermomechanical model is established using theories from the critical state and fractional plasticity. Based on the experimental observations, the shear and critical state characteristics of coarse-grained soils are dramatically affected by temperature variation. By introducing a thermal-dependent parameter, the temperature-dependent CSL can characterize the phenomenon in which the CSL moves approximately parallel downward at elevated temperatures. The direction of the non-orthogonal plastic flow is directly obtained by the fractional derivative of the modified elliptical YF, which provides a new approach for describing the non-orthogonality between the incremental direction of plastic strain and the yield surface. Subsequently, a thermomechanical model is established by combining the thermal state void-ratio-pressure parameter and non-orthogonal plastic flow rule. To evaluate the validity and capability of the proposed model, experimental data for Fujian sand, SPR, and crushed tuff were selected from the literature. Comparisons between the predictions and experimental data

show that the established thermomechanical model can reasonably reflect the drained shear characteristics of coarse-grained soils under different temperatures, including strain-hardening, strain-softening, dilatancy, contraction, and thermal softening.

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Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

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